The Last Mile Problem: Turning Executive Support into Real Open Source Contributions

Lessons from 10+ Years of AGL Activities

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OpenChain and Friends 2026 · Stuttgart

Who is This Guy?

- AGL Development and Community Manager since 2014
- Over 30 years embedded software experience
- Involved with Linux for 20 years
- Previously at MontaVista and Mentor Graphics
- Continental AG
- Motorola Mobile Devices
- Motorola Telematics Communications Group
- Defense Electronics at ITT Avionics and Northrop Grumman

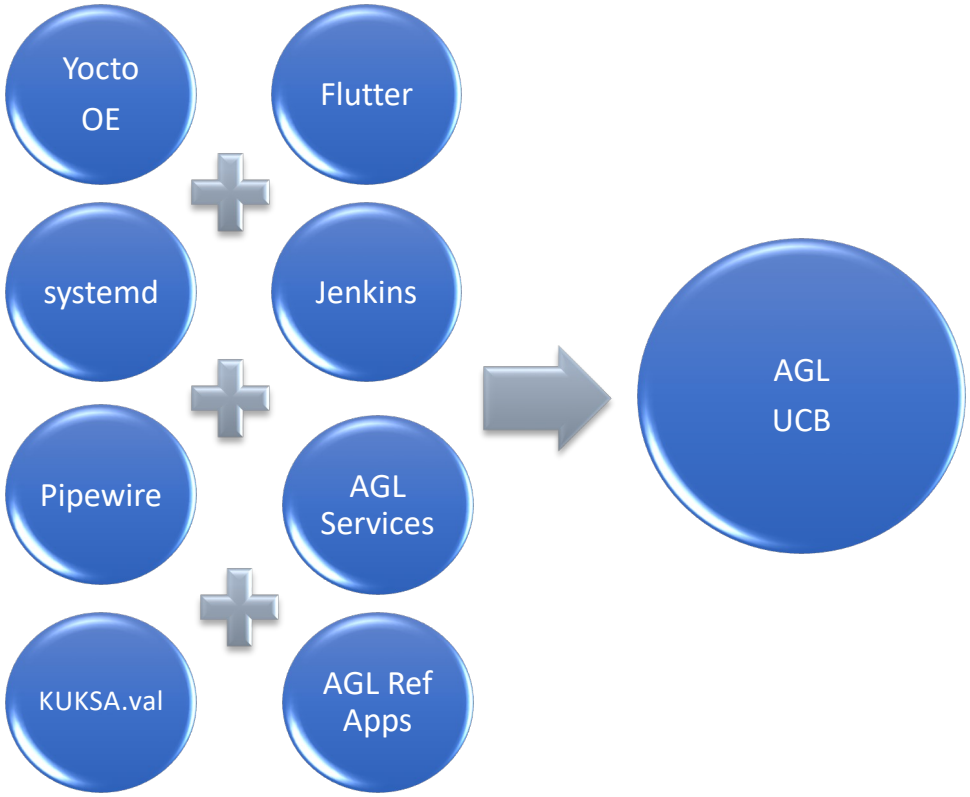
What is AGL?

- Non-profit organization
- Open source Linux-based collaborative project
- Hosted at Linux Foundation
- Collaborating to build the car of the future through *rapid innovation* by uniting the automotive and software industries

AGL at a Glance

- Linux Foundation collaborative project
- Current members include automakers, Tier 1s, and technology companies
- Unified Code Base (UCB) - Open platform started in 2015
- SoDeV – Open source reference plaform for Software-Defined Vehicles (SDVs)
- Production deployments: Toyota and Lexus from 2018 with a refresh in 2026, Subaru, Mazda, Mercedes-Benz Vans
- Expanded from IVI to instrument cluster, telematics, ADAS, and beyond

AGL Distro "Unified Code Base"



- *Unifying the best of Open Source into a single shared code base for the entire industry!*
- *Reduce fragmentation, focus on innovation and new features!*
- *Completely open source and transparent!*
- *Fully customizable; supported by an ecosystem of Tier Ones and service providers*

10+ Years of Tier 1/OEM Collaboration

- AGL proved that competitors can collaborate on shared software
- Neutral ground: OEMs and Tier 1s working side by side on a common platform
- Code contributions from automotive companies including Toyota, Honda, Panasonic, Aisin, Denso, Jaguar Land Rover, Denso Ten, Mitsubishi, Daimler and Suburu
- Shared investment reduces duplication and accelerates innovation
- Expert Groups enable focused, cross-company engineering work
- Result: production-ready open source software in millions of vehicles

2026 Toyota RAV4

Next generation of Toyota UI using Automotive Grade Linux and Flutter

Considerable investment in Flutter Embedder

https://pressroom.toyota.com/the-latest-evolution-of-toyotas-multimedia-coming-to-a-screen-near-you/

Images copyright Toyota

Suzuki e Vitara

- New Suzuki EVs feature AGL and QT
- Developed by Aisin and Yazaki

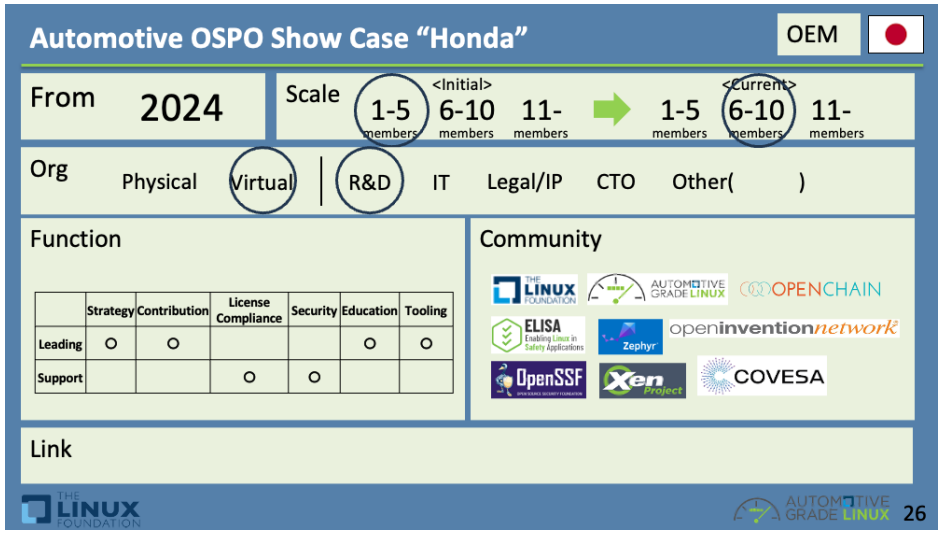
https://www.qt.io/press/suzuki-chooses-qt-to-power-its-digital-cockpit-for-mainstream-evs?hsLang=en

AGL OSPO Expert Group

- Launched November 2024 — led by Toyota
- Key members include Panasonic Honda
  - Encourage companies to establish their own OSPO
  - Share pain points and collaborate on solutions
  - Develop best practices for open source in automotive
  - Address business restrictions (export control, anti-trust)
  - Meets monthly — open to everyone!

AGL OSPO Expert Group

- Recognized the need to have open source sponsored at the highest levels of the company
- Created an Executive Slide Deck – Available for anyone to use to promote the value and usage of open to executives at your company!
  - Includes case studies from Honda, Toyota, Bosch and an unnamed Tier One



AGL OSPO Expert Group

- Many developers have concerns about contributing to open source projects
  - People will not be nice to me
  - My change is small, no one will want it
  - How do I contribute a patch?
- Presentation “Illusions and Truths of Participating in OSS Communites” targeted at developers
- English and Japanese versions available.

OSPO EXPERT GROUP SURVEY

WHAT TIER 1S AND OEMS TOLD US

OSPO EG Survey

- OSPO EG wanted to know what barriers are encountered by developers at automotive companies when contributing to open source
- Ten question areas
- English, Japanese, and Chinese language versions created
- 69 responses
  - 25 English, 37 Japanese, 8 Chinese

OSPO EG Survey

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Questions
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 1 Do you feel that you lack the necessary skills to contribute to open source software?
 2 Do you feel that you lack the knowledge about how to contribute to open source software?
 3 Are there language barriers that prevent you from contributing to open source software?
 4 Are there time constraints in your job that prevent you from contributing to open source software?
 5 Are there cultural issues affecting open source contributions?
6a Are your company's the rules and processes for contributions well-defined?
6b Are your company's rules and processes too complicated or difficult to follow?
 7 Are there issues with open source practices within your department?
 8 Are there issues with management understanding open source software?
 9 Are there IT system constraints in your company that make it difficult to access or contribute to open s
ource?
10 Are there budget constraints that prevent you contributing to open source ?
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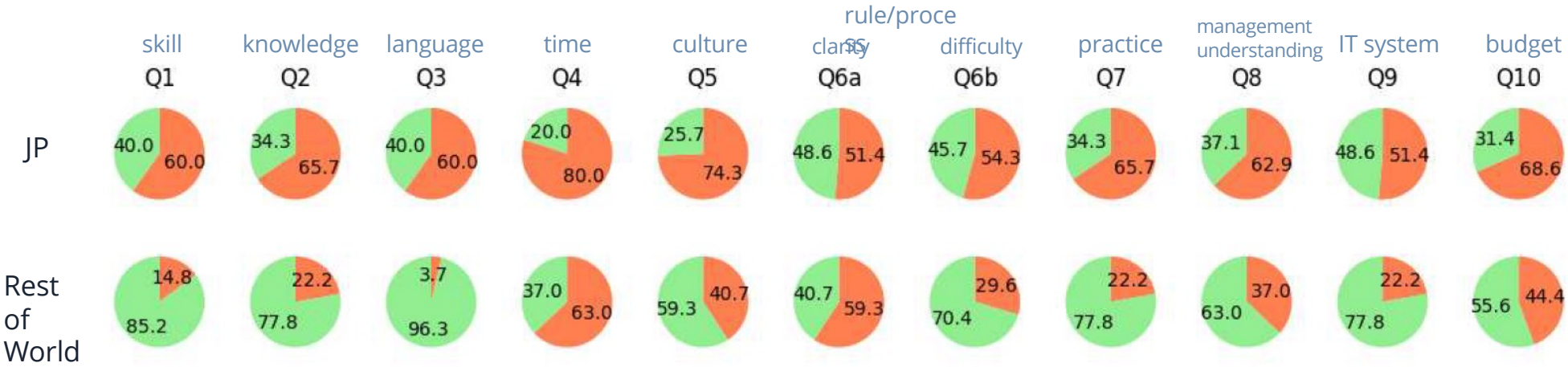
Survey results: JP vs ROW



Red(YES): Having barriers
Green(NO): Not having barriers

- Non-Japanese respondents more confident in their ability to contribute to OSS projects.
- Japanese respondents more likely to have language, skill, and company culture issues.
- Time and budget constraints across the board

Finding #1: Senior Management Buy-In

The #1 organizational barrier to open source contribution:

- Leadership doesn't see the business value of contributing
- Open source not seen as a strategic asset
- Concerns about competitive advantage for code that should be a commodity
- IP leakage concerns
- Without executive sponsorship, OSPO and contribution efforts stall

Result: Developers are blocked from posting code to open repositories

Finding #2: Developers Don't Know What to Do

Most common response from Tier 1s and OEMs:

- Developers lack clarity on internal open source policies
- No clear process for contributing code upstream to existing projects
- Process for creating new open source projects from internal work are undefined or complicated
- Fear of legal or IP consequences deters contribution

Result: Talented engineers want to contribute but don't know how

Breaking the Cycle: What We Can Do

- **Executive Champion – most important step! Majority of open source efforts will die without executive sponsorship**
- **Establish an OSPO** — a single place to go for managers and developers alike on how to use and contribute to open source projects
- **Create clear contribution guidelines** developers can follow
- **Educate management** — open source reduces cost and attracts talent
- **Use AGL as a neutral, low-risk space** to start contributing
- **Share success stories** across the OSPO Expert Group and industry gatherings
- **Leverage OpenChain standards** for compliance confidence

Call to Action

- OpenChain Automotive Working Group Monthly Call – “Automotive Open Source Governance”
- Join the AGL OSPO EG monthly call

Thank You
Join the AGL OSPO Expert Group

automotivelinux.org

agl-ospo-eg@lists.automotivelinux.org

Biweekly meetings — open to everyone!